Matching Matrices: From Feynman Diagrams to Qubits

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August 2026

Abstract

We study the change-of-basis matrices between orthogonal polynomials and their associated orthonormal functions — the eigenfunctions of a symmetry operator — in both the continuous and discrete settings.

In the continuous case, the Hermite functions (eigenfunctions of the Fourier transform) projected onto the Hermite polynomial basis yield an infinite matrix whose entries, after rescaling by $3^{(m+k)/2}$, are integers D_{mk} that count signed bipartite matchings — equivalently, Feynman diagrams. The generating function is $\sqrt{2\pi/3}\,\exp(-(s^2-4su+u^2)/3)$.

In the discrete case, the Krawtchouk polynomials (eigenvectors of the Walsh–Hadamard transform on n qubits) evaluated against a binomial weight at parameter q yield a finite matrix $I(m,k,n,q)$ whose generating function is $[1+\alpha(s+u)+su]^n$, where $\alpha = 1-2q$. For integer α, every entry is automatically an integer — a weighted count of multinomial microstates.

The two theories are connected by a scaling limit: as $n\to\infty$, the binomial converges to the Gaussian, Krawtchouk converges to Hermite, and multinomial coefficients converge to matchings. The passage from the discrete matrix to the continuous matrix is the passage from an n-qubit spin system to quantum field theory.

1 The Construction

Both cases follow the same pattern, in three steps.

Step 1: the polynomials. Fix a weight $W \geq 0$ on a set X. It defines an inner product on polynomials, $\langle f,g\rangle_W = \int_X f\,g\,W$. Gram–Schmidt produces the orthogonal polynomials P_n: $\langle P_m, P_n\rangle_W = h_n\delta_{mn}$.

Step 2: the functions. Multiplying by $\sqrt{W}$ turns them into the *associated functions* $\Phi_n = \widehat{P}_n\sqrt{W}$, which are orthonormal for the plain inner product: $\int_X \Phi_m\,\Phi_n = \delta_{mn}$.

Step 3: the coefficients. Project Φ_m back onto $\{P_k\}$ in $L^2(W)$:

$$b_{mk} = \langle \Phi_m, \widehat{P}_k\rangle_W = \int_X \widehat{P}_m\,\widehat{P}_k\,W^{3/2}.$$

The factor $W^{3/2}$ arises from $\sqrt{W}$ (from Φ_m) times W (from the inner product). The matrix (b_{mk}) is our object of study. The question: is there a rescaling that makes it integer, and if so, what do the integers count?

Part I: The Continuous Case

2 Hermite Polynomials and Hermite Functions

Let $H_n(t)$ denote the physicist's Hermite polynomials, defined by the recurrence

$$H_{n+1}(t) = 2t\,H_n(t) - 2n\,H_{n-1}(t), \qquad H_0(t) = 1, \quad H_1(t) = 2t,$$

and satisfying the orthogonality relation

$$\int_{-\infty}^{\infty} H_m(t)\,H_k(t)\,e^{-t^2}\,dt = 2^k\,k!\,\sqrt{\pi}\,\delta_{mk}. \tag{1}$$

Here $W = e^{-t^2}$ on $\mathbb{R}$, with $h_n = 2^n n!\sqrt{\pi}$. The *Hermite functions* are

$$\psi_m(t) = c_m\,H_m(t)\,e^{-t^2/2}, \qquad c_m = (2^m\,m!\,\sqrt{\pi})^{-1/2}. \tag{2}$$

These form an orthonormal basis for $L^2(\mathbb{R})$ and are eigenfunctions of the Fourier transform with eigenvalues $(-i)^m$.

> The Fourier transform $\widehat{f}(\xi) = \int f(t)\,e^{-2\pi i t\xi}\,dt$ is an operator of order four: $\widehat{\widehat{\widehat{\widehat{f}}}} = f$. Its eigenvalues must therefore be fourth roots of unity: $\{1, -i, -1, i\}$. The Hermite functions are the eigenfunctions, partitioned into four eigenspaces according to m mod 4. The fact that ψ_m is a Fourier eigenfunction is the statement that the quantum harmonic oscillator is invariant under the exchange of position and momentum: $x \leftrightarrow p$.

The L^2-projection of ψ_m onto the Hermite polynomial basis gives coefficients

$$a_{mk} = \frac{c_m}{2^k\,k!\,\sqrt{\pi}} \int_{-\infty}^{\infty} H_m(t)\,H_k(t)\,e^{-3t^2/2}\,dt. \tag{3}$$

The factor $e^{-3t^2/2}$ arises from $e^{-t^2/2}$ (from ψ_m) times e^{-t^2} (from the weight).

3 The Hermite Generating Function

Proposition 3.1. *The integrals $I_{mk} = \int_{-\infty}^{\infty} H_m(t)\,H_k(t)\,e^{-3t^2/2}\,dt$ have the generating function*

$$\sum_{m,k\geq 0} \frac{s^m\,u^k}{m!\,k!}\,I_{mk} = \sqrt{\frac{2\pi}{3}}\,\exp\!\Big(-\frac{s^2 - 4su + u^2}{3}\Big).$$

Proof. Using the generating function for Hermite polynomials, $\sum_{n\geq 0} H_n(t)\,z^n/n! = e^{2tz - z^2}$:

$$\begin{aligned}
\sum_{m,k} \frac{s^m u^k}{m!\,k!}\,I_{mk} &= \int_{-\infty}^{\infty} e^{2ts - s^2}\,e^{2tu - u^2}\,e^{-3t^2/2}\,dt \\
&= \int_{-\infty}^{\infty} \exp\!\big(-\tfrac{3}{2}t^2 + 2(s+u)t - s^2 - u^2\big)\,dt \\
&= \sqrt{\frac{2\pi}{3}}\,\exp\!\Big(\frac{2(s+u)^2}{3} - s^2 - u^2\Big) = \sqrt{\frac{2\pi}{3}}\,\exp\!\Big(-\frac{s^2 - 4su + u^2}{3}\Big).
\end{aligned}$$

□

Corollary 3.2 (Parity)**.** *$I_{mk} = 0$ whenever $m + k$ is odd.*

Proof. The exponent $-(s^2 - 4su + u^2)/3$ contains only terms of even total degree in (s, u). □

4 The Hermite Integer Matrix

Theorem 4.1 (Hermite integrality)**.** *Define $D_{mk} = 3^{(m+k)/2} \cdot I_{mk}/\sqrt{2\pi/3}$. Then $D_{mk} \in \mathbb{Z}$ for all $m, k \geq 0$ with $m+k$ even (and zero when $m+k$ is odd). Explicitly, with $b = (m-a)/2$, $c = (k-a)/2$,*

$$D_{mk} = \sum_{a} 4^a \, (-2)^{b+c} \, a! \, \frac{m!}{a!\, b!\, 2^b} \, \frac{k!}{a!\, c!\, 2^c} \,,$$

where each factor $n!/(a!\, b!\, 2^b)$ counts the partial matchings of $[n]$ with a unmatched points and b pairs.

Proof. Write $\exp(-(s^2 - 4su + u^2)/3) = \exp(-s^2/3)\, \exp(4su/3)\, \exp(-u^2/3)$. The coefficient of $s^m u^k$ is a sum over triples (a, b, c) with $m = 2b + a$ and $k = a + 2c$:

$$[s^m u^k] = \sum_{a,b,c} \frac{(-1/3)^b}{b!} \, \frac{(4/3)^a}{a!} \, \frac{(-1/3)^c}{c!}.$$

Multiplying by $m!\, k!\, 3^{(m+k)/2}$ absorbs all powers of 3 and leaves integer factors. □

Conjecture 4.2 (Hermite matching conjecture)**.** *D_{mk} counts signed bipartite matchings. Colour m points red and k points blue; then*

$$D_{mk} = \sum_{\substack{\text{perfect matchings} \\ \text{of the } m+k \text{ points}}} 4^{\#\{\text{red–blue pairs}\}} \, (-2)^{\#\{\text{same-colour pairs}\}}.$$

Equivalently,

$$D_{mk} = \sum_{\substack{c \geq 0 \\ m-c,\, k-c \text{ even}}} \binom{m}{c} \binom{k}{c} \, c! \, (m-c-1)!! \, (k-c-1)!! \, 4^c \, (-2)^{(m+k-2c)/2}.$$

Verified for all $m, k \leq 20$ (441 entries, zero discrepancies). The matching interpretation is consistent with Wick's theorem applied to $\mathbb{E}[H_m(X)\, H_k(X)]$ for $X \sim N(0, 1/3)$.

4.1 The diagonal

Theorem 4.3 (Hermite diagonal)**.**

$$\frac{D_{nn}}{n!} = \sum_{j=0}^{\lfloor n/2 \rfloor} \binom{n}{2j} \binom{2j}{j} \, 4^{n-2j} = 1,\ 4,\ 18,\ 88,\ 454,\ 2424,\ \ldots \quad (\textit{OEIS A081671}), \tag{4}$$

with ordinary generating function $1/\sqrt{(1-2x)(1-6x)}$ and exponential generating function $e^{4x}\, I_0(2x)$, where I_0 is the modified Bessel function of the first kind.

Proof. Setting $m = k = n$ in the matching formula and simplifying with $(2j-1)!! = (2j)!/(2^j\, j!)$ yields the stated sum. The OGF follows from $\sum_n a_n x^n$ using the Cauchy product of $1/\sqrt{1-4z^2}$ with $1/(1-4x)$, giving $1/\sqrt{(1-4x)^2 - 4x^2} = 1/\sqrt{1 - 8x + 12x^2}$. □

4.2 Edge entries

Proposition 4.4 (Hermite edge)**.** $D_{0,2n} = (-1)^n\,(2n)!/n! = (-1)^n\,2^n\,(2n-1)!!$.

4.3 Reverse direction

The *reverse* expansion $\widehat{H}_m = \sum_k \widetilde{b}_{mk}\,\psi_k$ has one power of $\sqrt{W}$ in place of three: $\widetilde{b}_{mk} = \int \widehat{H}_m\,\widehat{H}_k\,e^{-t^2/2}\,dt$. The same rescaling without the power of 3 gives

Theorem 4.5 (Hermite reverse)**.** $S_{mk} = (-1)^{(k-m)/2}\,D_{mk}$ *— the forward matrix with the signs stripped:*

$$S_{mk} = \sum_{\substack{\text{perfect matchings}\\ \text{of the } m+k \text{ points}}} 4^{\#\{\text{red–blue}\}}\;2^{\#\{\text{same-colour}\}}.$$

Every entry is non-negative.

4.4 Computed values

D_{mk} for even m and k, $0 \le m, k \le 14$

$m\backslash k$	0	2	4	6	8	10
0	1	−2	12	−120	1680	−30240
2	−2	36	−408	6000	−110880	2479680
4	12	−408	10896	−254880	6471360	−184222080
6	−120	6000	−254880	9529920	−339534720	12515731200
8	1680	−110880	6471360	−339534720	16590147840	
10	−30240	2479680	−184222080	12515731200		

Part II: The Discrete Case

5 Krawtchouk Polynomials

The *Krawtchouk polynomials* $K_j(x;n)$ for $p = \frac{1}{2}$ are defined on $\{0, 1, \dots, n\}$ by

$$K_j(x;n) = \sum_{r=0}^{j} (-1)^r \binom{x}{r}\binom{n-x}{j-r}, \tag{5}$$

with generating function

$$\sum_{j=0}^{n} K_j(x;n)\,z^j = (1-z)^x\,(1+z)^{n-x}, \tag{6}$$

and orthogonality

$$\sum_{x=0}^{n} K_j(x;n)\,K_k(x;n)\,\frac{\binom{n}{x}}{2^n} = \binom{n}{k}\,\delta_{jk}. \tag{7}$$

The Krawtchouk polynomials are the discrete Hermite polynomials: they are orthogonal with respect to the binomial distribution (the discrete Gaussian), and they converge to Hermite polynomials in the $n \to \infty$ limit. The matrix $(K_j(k;n))_{j,k=0}^{n}$ is, up to normalisation, the Walsh–Hadamard transform — the Fourier transform of $(\mathbb{Z}/2\mathbb{Z})^n$, i.e. the Hadamard gate applied to n qubits.

6 The Krawtchouk Matching Matrix

Definition 6.1 (Krawtchouk matching matrix)**.** Fix $n \geq 1$ and a parameter q. The *Krawtchouk matching matrix* is

$$I(m,k,n,q) = \sum_{x=0}^{n} K_m(x;n)\, K_k(x;n) \binom{n}{x} q^x\,(1-q)^{n-x} = \mathbb{E}\big[K_m(X;n)\, K_k(X;n)\big] \tag{8}$$

for $X \sim \mathrm{Bin}(n,q)$.

At $q = \frac{1}{2}$ this reduces to the orthogonality relation (7): the matrix is diagonal. For $q \neq \frac{1}{2}$ the off-diagonal entries are nonzero.

7 The Krawtchouk Generating Function

Theorem 7.1 (Bivariate generating function)**.** *Set $\alpha = 1-2q$. Then*

$$F(s,u) = \sum_{m,k=0}^{n} I(m,k,n,q)\, s^m\, u^k = \big[\,1 + \alpha(s+u) + su\,\big]^n. \tag{9}$$

Proof. Using the Krawtchouk generating function (6):

$$\begin{aligned} F(s,u) &= \sum_{x=0}^{n} \binom{n}{x} q^x\,(1{-}q)^{n-x}\,(1{-}s)^x(1{+}s)^{n-x}\,(1{-}u)^x(1{+}u)^{n-x} \\ &= \big[\,q(1{-}s)(1{-}u) + (1{-}q)(1{+}s)(1{+}u)\,\big]^n = \big[\,1 + \alpha(s{+}u) + su\,\big]^n. \end{aligned}$$

□

Compare the two generating functions:

	Continuous (Hermite)	Discrete (Krawtchouk)
GF	$\sqrt{\frac{2\pi}{3}}\,\exp\!\big(-\frac{s^2-4su+u^2}{3}\big)$	$[1+\alpha(s{+}u)+su]^n$
Shape	exp of quadratic form	n-th power
Parameter	$\sigma^2 = 1/3$ (fixed)	$\alpha \in \mathbb{Z}$ (free)

The continuous generating function is an *exponential* of a quadratic form; the discrete is an n-th *power* of an affine–bilinear form. They are connected by the usual $\exp = \lim(1+x/n)^n$: scaling s, u, α by $1/\sqrt{n}$ and sending $n \to \infty$, the power converges to the exponential.

8 Coefficient Extraction

Theorem 8.1 (Multinomial formula)**.**

$$I(m,k,n,q) = \sum_{c=\max(0,m+k-n)}^{\min(m,k)} \frac{n!}{(n-m-k+c)!\,(m-c)!\,(k-c)!\,c!}\,\alpha^{\,m+k-2c}. \tag{10}$$

Proof. Expand $[1+\alpha(s+u)+su]^n$ by the multinomial theorem with three terms. Pick a copies of 1, b copies of $\alpha(s+u)$, and c copies of su. Then $a+b+c=n$, and the monomial $s^m u^k$ requires $b=m+k-2c$ and $a=n-m-k+c$. □

Corollary 8.2 (Automatic integrality)**.** *When $\alpha \in \mathbb{Z}$, every entry $I(m,k,n,q) \in \mathbb{Z}$. No rescaling is required.*

9 Properties

Proposition 9.1 (Symmetries)**.**

1. ***Transpose:*** $I(m,k) = I(k,m)$.
2. ***Complement:*** $I(m,k,n,q) = \alpha^n\, I(n-m,\, n-k,\, n,\, q)$.
3. ***Parameter reflection:*** $I(m,k,n,1-q) = (-1)^{m+k}\, I(m,k,n,q)$.
4. ***No parity vanishing:*** *entries with $m+k$ odd are generically nonzero (unlike Hermite).*

Proposition 9.2 (Edge entries)**.** $I(0,k,n,q) = \binom{n}{k}\,\alpha^k$.

Proposition 9.3 (Special values)**.**

1. $\alpha = 0$ *($q=\frac{1}{2}$):* $F = (1+su)^n$, $I(m,k) = \binom{n}{k}\,\delta_{mk}$ *(orthogonality).*
2. $\alpha = 1$ *($q=0$):* $F = [(1+s)(1+u)]^n$, $I(m,k) = \binom{n}{m}\binom{n}{k}$.
3. $\alpha = -1$ *($q=1$):* $F = [(1-s)(1-u)]^n$, $I(m,k) = (-1)^{m+k}\binom{n}{m}\binom{n}{k}$.

Proposition 9.4 (Rational rescaling)**.** *If $\alpha = p/d$ with $p, d \in \mathbb{Z}$, then $d^{m+k}\, I(m,k,n,q) \in \mathbb{Z}$.*

9.1 Computed values

$I(m,k,n,q)$ for $\alpha=2$, $n=6$

$m\backslash k$	0	1	2	3	4	5	6
0	1	12	60	160	240	192	64
1	12	126	540	1200	1440	864	192
2	60	540	1935	3480	3240	1440	240
3	160	1200	3480	4900	3480	1200	160
4	240	1440	3240	3480	1935	540	60
5	192	864	1440	1200	540	126	12
6	64	192	240	160	60	12	1

$I(m,k,n,q)$ for $\alpha=2$, $n=8$

$m\backslash k$	0	1	2	3	4	5	6	7	8
0	1	16	112	448	1120	1792	1792	1024	256
1	16	232	1456	5152	11200	15232	12544	5632	1024
2	112	1456	8092	24976	46480	52864	35392	12544	1792
3	448	5152	24976	66136	103600	97216	52864	15232	1792
4	1120	11200	46480	103600	134470	103600	46480	11200	1120
5	1792	15232	52864	97216	103600	66136	24976	5152	448
6	1792	12544	35392	52864	46480	24976	8092	1456	112
7	1024	5632	12544	15232	11200	5152	1456	232	16
8	256	1024	1792	1792	1120	448	112	16	1

$I(m,k,n,q)$ for $\alpha=3$, $n=6$

$m\backslash k$	0	1	2	3	4	5	6
0	1	18	135	540	1215	1458	729
1	18	276	1710	5400	8910	6804	1458
2	135	1710	8385	19620	21465	8910	1215
3	540	5400	19620	30800	19620	5400	540
4	1215	8910	21465	19620	8385	1710	135
5	1458	6804	8910	5400	1710	276	18
6	729	1458	1215	540	135	18	1

Part III: Comparison and Physics

10 Structural Comparison

Feature	Continuous (Hermite)	Discrete (Krawtchouk)
Weight	e^{-t^2} on $\mathbb{R}$	$\binom{n}{x}/2^n$ on $\{0,\dots,n\}$
Polynomials	$H_n(t)$	$K_j(x;n)$
Functions	$\psi_n = c_n H_n e^{-t^2/2}$	(absorbed into weight)
Transform	Fourier on $\mathbb{R}$	Walsh–Hadamard on $\{0,1\}^n$
Eigenvalues	$(-i)^n$	$(-1)^j$
GF	$\sqrt{\frac{2\pi}{3}}\, e^{-(s^2-4su+u^2)/3}$	$[1+\alpha(s+u)+su]^n$
Integrality	rescale by $3^{(m+k)/2}$	automatic for $\alpha \in \mathbb{Z}$
Parity	$D_{mk}=0$ for $m+k$ odd	no vanishing
Edge	$(-1)^n(2n)!/n!$	$\binom{n}{k}\alpha^k$
At $\alpha=1$	—	$\binom{n}{m}\binom{n}{k}$
Symmetry	$(m,k)\leftrightarrow(k,m)$	transpose + complement
Combinatorics	signed matchings	multinomial coefficients
Physics	Feynman diagrams	qubit microstates

11 Quantum Mechanics

The matching matrices are not merely combinatorial curiosities. They sit at the centre of quantum mechanics, and the passage from the discrete matrix to the continuous matrix is the passage from finite spin systems to quantum field theory.

11.1 The harmonic oscillator

The Hermite functions ψ_n are the energy eigenstates of the **quantum harmonic oscillator**, the single most important quantum system. Its Hamiltonian is $H = p^2 + x^2$ (up to constants), with eigenvalues $E_n = (n+\frac{1}{2})\hbar\omega$. Every quadratic quantum system — photons, phonons, the free electromagnetic field — is a harmonic oscillator.

Observation 11.1 (Physical meaning of D_{mk})**.** The Hermite matching integer D_{mk} is proportional to the overlap between the m-th energy eigenstate (ψ_m) and the k-th polynomial observable (H_k) in $L^2(e^{-t^2})$. Physically: if a particle is in the m-th energy level, D_{mk} measures the k-th polynomial moment of its position, weighted by the Gaussian ground state.

11.2 Matchings are Feynman diagrams

Wick's theorem states that the expectation of a product of Gaussian random variables decomposes into a sum over all pairings:

$$\mathbb{E}[X_1 X_2 \cdots X_{2n}] = \sum_{\text{matchings}} \prod_{\text{pairs } (i,j)} \mathbb{E}[X_i X_j].$$

Each matching is a **Feynman diagram**: a way of contracting field operators into propagators. In free quantum field theory, every scattering amplitude is a sum over such diagrams.
The Hermite matching integer D_{mk} — the sum over bipartite matchings with weights $+4$ (cross-pairs) and -2 (self-pairs) — *is* Wick's theorem applied to $H_m(X)\,H_k(X)$ at variance $\sigma^2 = 1/3$:

- A **cross-pair** connects a slot from H_m to one from H_k: an inter-level transition, weighted $+4$.
- A **self-pair** connects two slots within the same polynomial: an intra-level contraction, weighted -2.

The Hermite matching matrix is a **table of Feynman diagram counts**, indexed by energy levels.

11.3 Qubits and spin

The Krawtchouk matching matrix has its own quantum mechanical meaning.
Krawtchouk polynomials are the matrix elements of $SU(2)$ representations. A system of n spin-$\frac{1}{2}$ particles (n qubits) has a Hilbert space of dimension 2^n. The total spin operator has eigenspaces labelled by $x \in \{0, \ldots, n\}$ (the number of spin-ups), and $K_j(x; n)$ is the j-th matrix element.
The Walsh–Hadamard transform ($K_j(k; n)$) is the Hadamard gate applied to n qubits — the first gate in Deutsch–Jozsa, Grover, and Shor.

Observation 11.2 (Physical meaning of $I(m, k, n, q)$)**.** $I(m, k, n, q)$ measures the correlation between the m-th and k-th Hadamard-basis observables when the n qubits are in a biased product state with probability q of spin-up.
At $q = \frac{1}{2}$ ($\alpha = 0$): the state is $|+\rangle^{\otimes n}$. The observables are decorrelated (orthogonality).
At $q \neq \frac{1}{2}$: the observables become correlated. The parameter α measures the strength of quantum correlation.

11.4 The generating function as a partition function

The generating function $[1 + \alpha(s{+}u) + su]^n$ is a partition function. Each of the n qubits independently contributes one of three terms:

Term	Weight	Meaning
1	1	qubit idle
αs or αu	α	qubit contributes to one observable
su	1	qubit contributes to both

This is a lattice model: n independent sites, each in one of three effective states, with a global constraint (total s-degree $= m$, total u-degree $= k$).

11.5 Why integers?

Observation 11.3 (Integrality as discreteness of process space)**.** The integers count the number of topologically distinct quantum processes.

- **Continuous:** D_{mk} counts signed Feynman diagrams — the distinct ways energy levels m and k can exchange virtual quanta through Gaussian contractions. The integrality says that the space of quantum processes has a discrete structure, even though the underlying physics is continuous.
- **Discrete:** $I(m, k, n, q)$ counts weighted microstates — the distinct ways n spins can collectively produce a given pair of observables. The integrality is automatic because we are counting configurations of a finite system.

11.6 From spins to fields

The passage from the discrete to the continuous matrix is the passage from statistical mechanics to quantum field theory:

$$\underbrace{I(m,k,n,q)}_{n \text{ qubits}} \xrightarrow{n\to\infty} \underbrace{D_{mk}}_{\text{QHO / QFT}} .$$

In this limit:

- $\mathrm{Bin}(n, q) \to N(0, \sigma^2)$ (central limit theorem)
- Krawtchouk polynomials $\to$ Hermite polynomials
- multinomial coefficients $\to$ matchings (Wick contractions)
- lattice model $\to$ free field theory
- $[1 + \alpha(s+u) + su]^n \to \exp(-(s^2 - 4su + u^2)/3)$ ($\exp = \lim(1 + x/n)^n$)

The matching matrix is a Rosetta stone between these two descriptions. Every Feynman diagram is a limit of multinomial microstates.

11.7 Quantum error correction

Observation 11.4 (MacWilliams identity)**.** The complement symmetry $I(m,k,n,q) = \alpha^n\, I(n{-}m,\, n{-}k,\, n,\, q)$ is the MacWilliams identity of coding theory. For a quantum error-correcting code of length n, the Krawtchouk transform relates the weight distribution of the code to that of its dual. The matching matrix generalises this: it measures how a code's weight distribution transforms under a change of bias parameter.

12 Open Questions

Conjecture 12.1 (Discrete combinatorial interpretation)**.** *For integer $\alpha \geq 1$, the entry $I(m,k,n,q)$ counts something beyond multinomials: weighted lattice paths, coloured set partitions, or a discrete matching analogue. A bijective proof would illuminate the convergence to bipartite matchings as $n \to \infty$.*

Conjecture 12.2 (Discrete diagonal generating function)**.** *For fixed α, the diagonal sequence $I(k,k,n,q)$ should have a closed-form generating function generalising the Hermite diagonal's $1/\sqrt{(1{-}2x)(1{-}6x)}$.*

Conjecture 12.3 (Scaling limit)**.** *Under the substitutions $m = \lfloor \mu\sqrt{n}\,\rfloor$, $k = \lfloor \kappa\sqrt{n}\,\rfloor$, $\alpha = \beta/\sqrt{n}$, the discrete matrix converges to the continuous:*

$$\lim_{n\to\infty} I\big(\lfloor \mu\sqrt{n}\,\rfloor,\, \lfloor \kappa\sqrt{n}\,\rfloor,\, n,\, q\big) \sim D^{Hermite}_{\mu,\kappa} \cdot (\textit{normalisation}).$$

Conjecture 12.4 (Hermite matching conjecture, restated)**.** *A proof of Conjecture 4.2 via Isserlis' theorem (Wick's theorem at non-unit variance) would establish that D_{mk} is a sum over Feynman diagrams — not merely an analogy but an identity.*

13 Proofs Available Upon Request

All entries in the continuous tables were computed from the proved coefficient-extraction formula (Theorem 4.1) and independently verified against the matching formula (Conjecture 4.2) for all $m,k \leq 20$ (441 entries, zero discrepancies). All discrete tables were computed from the multinomial formula (Theorem 8.1) and independently verified against the direct Krawtchouk sum (8) for $n \leq 12$ and all $\alpha \in \{-3,-2,-1,0,1,2,3,\frac{1}{3},\frac{1}{2},\frac{2}{3},\frac{3}{2}\}$.